

# Ekaterina Antipushina

NeuroAI & Embodied AI · ML Research & Engineering

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## Research profile

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Researcher and machine-learning engineer working on multimodal learning for neural signals, medical images, and 3D scenes. Research spans representation learning, generative modeling, real-time neurofeedback, and spatial reasoning with vision-language models. PhD student at Skoltech and ML Engineer at the Spatial Intelligence Lab, Applied AI Institute.

## Research and engineering experience

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### Machine Learning Engineer

October 2025 – present

Spatial Intelligence Lab, Applied AI Institute

Developing and benchmarking LLM and vision-language systems for 3D scene understanding and language-guided object localization. Working with scene graphs, geometry, tool use, and evaluation of spatial reasoning in a joint project with Huawei.

### Research Engineer, Medical AI

November 2024 – September 2025

Skoltech · Neuroimaging and Cognitive Neuroscience Lab

Developed pyOpenNFT, an open-source Python framework for real-time fMRI and EEG-fMRI neurofeedback. Implemented parallel signal processing with shared memory and a FastAPI prediction service for integrating machine-learning models. First author of the MICCAI 2025 framework paper.

### Machine Learning Engineer · part-time

July 2024 – August 2025

BIMAI-Lab, University of Sharjah

Led computational analysis of transcriptomic and proteomic data in international studies of human brain organoids under hypoxia. Developed statistical and machine-learning workflows, interpreted results with the experimental team, and contributed to papers in Fluids and Barriers of the CNS and iScience.

### ML Engineer / Research Engineer

January 2023 – December 2024

Skoltech · Medical AI / BIMAI-Lab

Developed generative and interpretable models for multimodal neuroimaging and biomedical data, including Rest2Task, EEG-to-ECOG modeling, schizophrenia classification, and kidney-cancer metastasis prediction. Worked with data fusion, feature selection, graph representations, and validation on small medical datasets.

## Education

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**PhD student** · Skolkovo Institute of Science and Technology (Skoltech) · 2024 – present

**Master of Science** · Skoltech · 2022 – 2024

**Bachelor's degree in Biomedical Engineering** · Moscow Aviation Institute · 2018 – 2022

## Methods and tools

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Python, PyTorch, TensorFlow, scikit-learn, NumPy/pandas; Transformers, VAE/GAN, optimal transport; multimodal fusion, statistical modeling, subject-wise validation; FastAPI, multiprocessing, Docker, Git, Linux, GPU computing.

## Publications and preprints

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### Accepted papers · 2026

1. Antipushina E., Sukhorukov D., Cherkasskaya E., Makarov I. *Geometry-Aware Multi-View Cardiac MRI Segmentation and Biomarker Estimation*. 1st MICCAI Workshop on Medical World Models, 2026. [OpenReview](#).
2. Sukhorukov D., Antipushina E., Makarov I. *TABS: Transfer-Aware Backbone Selection for Cross-Centre Ultrasound Biometry*. 1st MICCAI Workshop on Medical World Models, 2026. [OpenReview](#).
3. Sukhorukov D., Antipushina E., Koush Y., Makarov I. *Speckle-Phase Refinement on an Exact Differentiable Scanner Twin for Inverse OCT Phantom Generation*. SynthOCT Challenge, MICCAI, 2026. [OpenReview](#).

### Published papers

1. Gaston-Breton R., Bouzid A., Antipushina E., et al. *Female metabolic resilience and male-biased protein quality defects under hypoxia in human brain organoids*. iScience 29(5), 115714, 2026. [10.1016/j.isci.2026.115714](#).
2. Kalimullin R., Antipushina E., Razorenova A., Kormakov G., Koshev N. *CSTNet: A Generative Framework for EEG-to-ECoG Mapping via Optimal Transport*. EMA4MICCAI 2025, Springer, pp. 154–162, 2026. [10.1007/978-3-032-13961-0\\_16](#).
3. Gaston-Breton R., Bouzid A., Antipushina E., et al. *Translational biomarkers of hypoxic brain injury uncovered in CSF secreting human choroid plexus organoids*. Fluids and Barriers of the CNS 22, 117, 2025. [10.1186/s12987-025-00731-z](#).
4. Antipushina E., Davydov N., De Feo R., et al. *pyOpenNFT: An Open-Source Python Framework for ML-Based Real-Time fMRI and EEG-fMRI Neurofeedback*. MICCAI 2025, Springer, 2025. [10.1007/978-3-032-05162-2\\_55](#).
5. Boyko M., Antipushina E., Bernstein A., et al. *Interpretable AI models for predicting distant metastasis development based on genetic data: Kidney cancer example*. BIO Web of Conferences 100, 01009, 2024. [10.1051/bioconf/202410001009](#).
6. Piliugin N., Knyshenko M., Sain A., Antipushina E., Soghoian G., Lebedev M. *Artificial Sensations Evoked by Transcutaneous Electrical Nerve Stimulation: Investigating the Parameters Space*. IEEE Ural-Siberian Conference on Computational Technologies in Cognitive Science, Genomics and Biomedicine (CSGB), 2023. [10.1109/CSGB60362.2023.10329845](#).
7. Antipushina E., Antipushina D. N. *Left Ventricular Assist Devices in the Management of End-Stage Heart Failure*. Conference paper, 2021. [Publication record](#).

### Preprints

1. De Feo R., Antipushina E., Tyukin I., Koush Y. *EEG-to-fMRI Prediction for Neurofeedback: Evaluating Regularized Regression and Clustering Approaches*. bioRxiv, 2025. [10.1101/2025.05.20.654907](#).
2. Antipushina E., Zubrikhina M., Kalimullin R., Kotoyants N., Sharaev M. *Comparative analysis of Pearson and Canonical correlation-based functional connectivity matrices for neuroimaging classification tasks*. bioRxiv, 2024. [10.1101/2024.04.23.590747](#).

## Research and software projects

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### 3D Spatial Understanding

Developing and benchmarking LLM and vision-language systems for 3D scene understanding and language-guided object localization. Working with scene graphs, geometry, tool use, and evaluation of spatial reasoning in a joint project with Huawei.

### Neural Representation Learning

Developed tokenization and pretraining methods for EEG and fMRI using Transformer/ViT architectures, BSQ and VQ-VAE representations, and masked modeling. Evaluated transfer across subjects and recording configurations with controls for data leakage.

### Medical Image Modeling

Research on cardiac MRI segmentation and biomarker estimation, transfer-aware ultrasound biometry, and inverse OCT phantom generation. Three accepted contributions to MICCAI 2026 workshops and challenges.

### Brain Organoid Data Analysis

Computational analysis of molecular responses to hypoxia in human brain and choroid plexus organoids. Statistical modeling and integration of omics data supported studies published in *Fluids and Barriers of the CNS* (2025) and *iScience* (2026).

### pyOpenNFT

Open-source Python framework for real-time fMRI and EEG-fMRI neurofeedback with machine-learning integration, parallel signal processing, and a prediction service.

### CSTNet

Generative EEG-to-ECOG mapping using optimal transport. Contributed to the framework described in the EMA4MICCAI 2025 paper, published by Springer in 2026.

### Rest2Task

Generative framework for predicting task-based fMRI from resting-state recordings using variational autoencoders and generative adversarial models.

### Neuroforum

Designed a platform for researchers and AI agents working with scientific materials. Implemented a FastAPI service, MCP tools, access controls, background jobs with RabbitMQ/Dramatiq, LLM cost accounting, Docker Compose, and CI checks.

## Posters

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Rest2Task: generative modeling for task-based fMRI prediction. Sber RnD Day, 2024.

The application of machine learning methods for multimodal neuroimaging data in the diagnosis of schizophrenia. LIFT Conference, 2024.

Development of a personalized Transcranial Magnetic Stimulation complex utilizing biofeedback. Poster, 2024.

Benchmarking of Machine Learning and Deep Learning approaches for Neuroimaging Data. Poster, 2023.

## Research talks

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Universal Brain Encoder: The Evolution of Models for Neural Signals. NeuroTalk, MISIS University, April 2026.

Bridge Between Time and Space: How Generative AI Turns EEG into fMRI. DataFest (ODS), March 2025.